

Conservation Genetics Homework #1

Name: _____

- 1) Explain why Hardy-Weinberg Equilibrium and Linkage Equilibrium can both be considered null models.
- 2) Ten individuals from a population of Great Gray Spotted Sedge (*Arctopsyche grandis*) is genotyped at two diallelic loci, *A* and *B*. In the resulting sample, five individuals are homozygous for the haplotype A_1B_1 and five individuals are homozygous for the haplotype A_2B_2 . Are *A* and *B* linked? Why or why not?
- 3) A graduate student in the Department of Ecology genotypes 400 Ferruginous Hawks in Oregon ("Pop 1") and 400 Ferruginous Hawks east of the Continental Divide in Montana ("Pop 2"). In Pop 1 they find 324 birds are homozygous recessive, 72 birds are heterozygous, and 4 birds are homozygous dominant. In Pop 2 they find 196 birds are homozygous recessive, 168 birds are heterozygous, and 36 birds are homozygous dominant. Determine the total number of *A* and *a* alleles in each subpopulation, as well as their relative frequencies. Next, pool the allele counts for Pop 1 and Pop 2 and do the same for the "global" population of hawks in the study (i.e., treat all hawks as belonging to the same population).

- a. What are the expected genotype frequencies of Pop 1 under HWE?

$$f(AA) = \underline{\hspace{2cm}} \qquad f(Aa) = \underline{\hspace{2cm}} \qquad f(aa) = \underline{\hspace{2cm}}$$

- b. What are the expected genotype frequencies of Pop 2 under HWE?

$$f(AA) = \underline{\hspace{2cm}} \qquad f(Aa) = \underline{\hspace{2cm}} \qquad f(aa) = \underline{\hspace{2cm}}$$

- c. What are the expected genotype frequencies of the global population under HWE?

$$f(AA) = \underline{\hspace{2cm}} \qquad f(Aa) = \underline{\hspace{2cm}} \qquad f(aa) = \underline{\hspace{2cm}}$$

- d. Is Pop 1 currently under HWE?* (YES / NO)
Is Pop 2 currently under HWE?* (YES / NO)
Is the global population currently in HWE?* (YES / NO)

* don't worry about performing a χ^2 test!